

To: Carleton University Simulator Project
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Introduction:

The purpose of this report is to give an outline for the tasks that will be performed by Olamide Olayinka in the academic year of 2015-2016 for the Carleton University Simulator Project(CUSP). As part of the structural team, Olamide Olayinka is going to focus on the hatch, particularly the external hatch components, in a sub-team of three other members. The hatch is an assembly of three main parts. These parts are the External hatch components, latch connections, and the hinge mounting assembly. Also the entry and egress bar would be designed. Each member has divided their tasks equally and the main focus is to have the hatch assembled. In this proposal, guidelines for the tasks to be completed and a rough estimated time line of their completion will be discussed.

CUSP Goals

The goal for this academic year is to have hatch completely fitted unto the sphere with all its components. The fully functioning Hatch needs to be completed and assembled with the sphere for the Atlas Simulator by April 2016 [1].

Hatch Current Status:

The hatch plates has been delivered by the manufacturer. Most of the parts needed for assembly have also been manufactured. One of the hatch ring has been manufactured and would need a second one [2].

Hatch Team:

The Hatch team consists of four members from the structures team. Three members of the hatch team would focus on hatch related components while one member would focus on

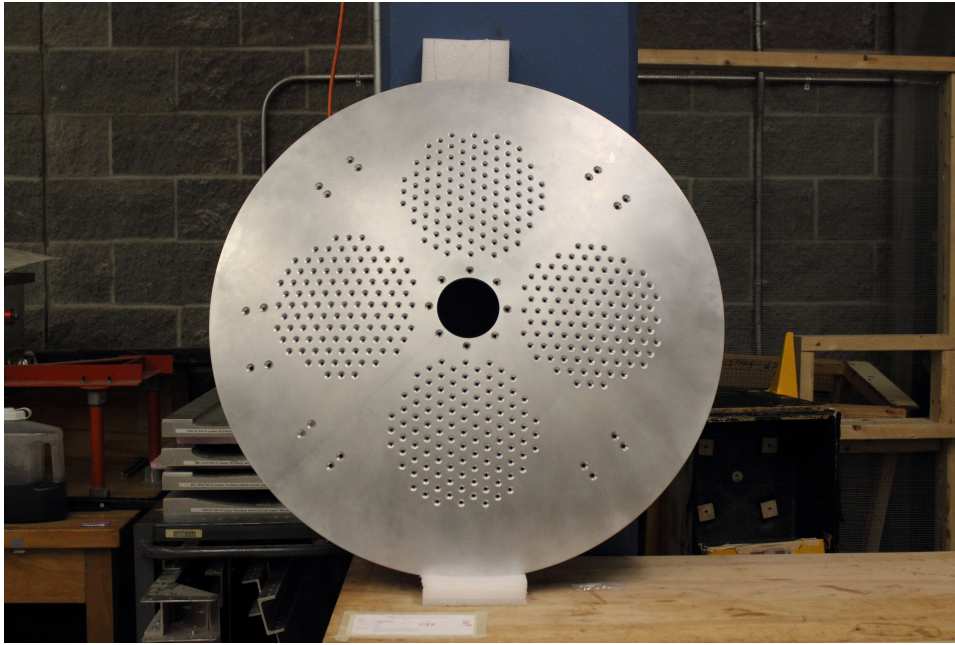


Figure 1: Hatch plate showing ventilation holes and mounting holes

designing the safety entry and egress bar.

Hatch Plate:

The design for the Hatch plates has been completed and two plates have been manufactured. All the basic calculations and finite element analysis will be revisited and revised if need be to ensure the safe operation of the hatch plates. In ANSYS 14.0, a contact patch of elliptical shape and area of 3.18 sq-in was used on the ventilation holes, where stress is most concentrated, to determine its ability to withstand the predicted 1500lb load from the mecanum wheels. The maximum stress of 25.9 ksi was calculated by the ANSYS FEA simulation. The yield stress of the material chosen, aluminium 6061-T6, is 40 ksi. This means the hatch plate has a safety factor of 1.6 [2]. Further analysis on the hatch would be performed in ANSYS to understand its performance when subjected to higher loads or different contact patch areas to account for future roller material and contact area adjustment that may be made.

Hatch Ring:

One of the hatch rings has been manufactured using 3D printing. Some of the ring segments are defective and hence some new ones might have to be printed. Also the rings will be tested under similar loading conditions using the MTS testing machine to ensure that it can withstand the loads of the mecanum wheels. The rings would eventually be glued on the hatch flange using epoxy. The material chosen for the rings would be re-examined to ensure it is well suited for this application

Hatch Flange:

The Hatch flange is currently installed on the sphere temporarily. Both Flanges would be removed from the sphere petals for further machining. A step down needs to be machined on flanges to allow for the resting of the hatch ring. Also four Slots for the latch slam bolts would be machined on the inner parts of the flange, all at 90° to each other [3].

Fans:

Eight fan baffles are to be installed on both hatch plates. The fans will be wired together so as to not have individual wires loitering around.

Activities:

- Verify all the calculations and run a finite element analysis on hatch plate using varied contact patch areas in ANSYS
- machine down extra weld material deposited on the hatch flange
- Machine a step down on both hatch flanges
- Assemble fans with hatch plate
- Manufacture second ring and test material selected

Timeline:

Table 1: Estimated Timeline for Project Objectives

Task	Sept.	Oct.	Nov.	Dec.
Complete FEA analysis to confirm hatch plate design	XX			
Machine a step down for hatch ring		XX		
Manufacture Both hatch rings			XX	
Machine latch connection points on to the flanges			XX	
Permanently glue hatch rings on to the hatch flange.				XX

References

- [1] CUSP. CUSP Manual 2014. Reference Manual, Carleton University, CUSP Manual 2014, 2014.
- [2] Joyoty Dasgupta. Hatch Components Design. Design Report, Carleton University. DR-STR-JD 15.HatchComponentsDesign.02
- [3] William McNair. Hatch Design.Design Report, Carleton University, DR-STR-wm.14.Hatch Design.01.2014